

Reconstructing the Goldberg–Palatini Action from Noether Identities

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Abstract

We investigate the reconstruction of the first-order gravitational action from the Noether identities associated with diffeomorphism invariance. The independent fields are a symmetric perturbation of the contravariant metric density and an independent trace-adjusted connection, with prescribed transformation laws. In generic spacetime dimension, we construct a local, Lorentz-covariant ansatz containing quadratic terms with at most two derivatives and a cubic interaction restricted to the structure ϕGG . The action kernels are expanded in tensor bases comprising 31 independent coefficients. Expanding the Noether identities order by order in the fields yields algebraic constraints, which we solve using symbolic computation. These constraints eliminate the $\phi\phi$ kernel and determine the remaining coefficients up to an overall normalization, recovering the Goldberg–Palatini action. Since the connection can be eliminated exactly, the result also determines the complete second-order action, with its infinite set of graviton vertices. The construction requires no choice of an energy–momentum tensor to guide the self-coupling procedure and directly identifies the restrictions imposed by the prescribed symmetry. The result establishes uniqueness within the assumed class of actions. All tensor bases, identities and solutions are provided in appendices, and the symbolic computations will be made available as supplementary material.

I. INTRODUCTION

The extent to which gravitational dynamics can be recovered from the consistency of an interacting massless spin-two field is a longstanding question. In the self-coupling construction, which goes back to Feynman [1] and was completed by Deser [2], the use of first-order variables allows the nonlinear completion to be expressed through a cubic interaction, avoiding the infinite expansion of the metric formulation. Discussions of this approach have also emphasized the assumptions entering the construction, the ambiguities in the energy–momentum tensor, and the treatment of boundary terms [3, 4]. These issues motivate a formulation in which the constraints imposed by gauge symmetry can be examined directly.

Local gauge invariance implies differential identities among the Euler–Lagrange derivatives of the action. These are identities of Noether’s second theorem and hold off shell. Once

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the field transformations and a class of admissible actions have been specified, they can be used as algebraic conditions on unknown interaction coefficients. This reverses the usual direction of the calculation: instead of deriving identities from a known action, one uses the identities to determine which actions are compatible with the prescribed transformations. This strategy was used in Ref. [5] to determine the scalar–graviton vertex in noncommutative space, where the closed form of the action is not available, from the invariance under a restricted class of coordinate transformations.

In this work we apply this strategy to the first-order gravitational variables $\phi^{\mu\nu}$ and $G^\lambda_{\mu\nu}$. The former is the perturbation of the contravariant metric density, while the latter is a trace-adjusted connection treated as an independent field. The same strategy applies in the metric formulation, where the field-dependent part of the transformation is essential: invariance under the linearized transformations alone fixes only part of each vertex, whereas the full transformation determines the three-graviton vertex from the quadratic term, up to total derivatives [6]. The programme does not terminate there, however, since new vertices appear at every order. The first-order choice is adapted to the polynomial structure of the Goldberg–Palatini action [7–9] and to its use in perturbative gravity [10]. We restrict the action to quadratic kernels of prescribed derivative order and to a derivative-free cubic interaction of the form ϕGG , expand the kernels in tensor bases, and impose the Noether identities order by order in the fields; the determination of the action then reduces to a finite system of linear constraints on 31 coefficients, which is solved completely.

The result is not confined to the first-order theory. The field equation of G is algebraic, so once the first-order action is fixed the connection can be eliminated exactly, for $d > 2$ and nondegenerate metric density: G is expressed through $\partial\phi$ by inverting the kernel of the GG term, which is linear in $\eta + \kappa\phi$, and substituting back gives the second-order action as a quadratic form in $\partial\phi$ whose kernel is that inverse, Eq. (3.13) of Ref. [10]. The formal perturbative expansion of the inverse, Eq. (3.18) there, then generates the complete infinite set of graviton self-interaction vertices of the metric formulation. The finite determination of the first-order action therefore fixes all of them.

Our objective is to determine the freedom remaining in this ansatz after the Noether identities are imposed. The transformation laws, the field content, the derivative bounds and the restriction on the cubic interaction are inputs to the construction. Accordingly, the uniqueness statement concerns this specified class of actions; it does not assert that diffeo-

morphism invariance alone selects Einstein gravity among all local gravitational theories.

The present construction complements the self-coupling approach of Refs. [2, 4] by determining the admissible action kernels directly from prescribed nonlinear gauge transformations. Its contribution is an explicit finite-dimensional uniqueness analysis within the stated ansatz, identifying which Noether identities remove each remaining freedom. The self-coupling procedure has recently been extended to metric-affine theories, with the connection as an independent field [11]; that analysis identifies the choice of energy–momentum tensor as the source of the ambiguities, notes that in the first-order construction of Ref. [2] the connection is an auxiliary field that is not itself bootstrapped, and observes that the existing arguments establish existence rather than uniqueness. The present construction, within the stated ansatz, addresses these three points: no energy–momentum tensor is chosen, which is the point at issue in Ref. [3], the geometrical input being retained explicitly in the transformation laws; the variations of both independent fields enter the Noether identity and are included in the determination; and uniqueness is established within the stated class. The idea of using the consistency of the gauge identities instead of self-coupling goes back to Wald [12], who analyzed the consistent nonlinear extensions through identities among the field equations; here the nonlinear transformations are prescribed and the compatible action is determined within a finite ansatz. The result concerns the bulk action modulo total derivatives and does not determine boundary terms.

The distinction can also be expressed in terms of Noether’s two theorems. Energy–momentum self-coupling constructions combine the conserved current associated with global translations with the consistency conditions imposed by the gauge identities of the spin-two field. Here we impose the off-shell identities of Noether’s second theorem directly, without constructing an energy–momentum tensor as an intermediate step. Translation symmetry is already contained in the prescribed transformations. The difference therefore lies in the organization and inputs of the procedure: the nonlinear transformations are specified at the outset, and the compatible action is determined within the chosen ansatz.

The paper is organized so that the main text carries the logic of the construction and the appendices carry its details. Section II fixes the fields and their transformations, Section III the ansatz and its counting, Section IV the identities and their decomposition into sectors, Section V the solution, and Section VI the checks and the scope of the result. Figure 1 summarizes the procedure. Explicit tensors, bases, identities and kernels are collected

in Appendices B–E, and Appendix F describes the symbolic implementation and how to reproduce every check.

II. FIRST-ORDER VARIABLES AND GAUGE TRANSFORMATIONS

a. Conventions. Units are $\hbar = c = 1$, so that dimensions are mass dimensions. We work in d spacetime dimensions with a flat metric $\eta_{\mu\nu}$ of signature $(+, -, \dots, -)$; the algebraic tensor identities used below are polynomial identities in η and the momenta and are independent of the signature, and for generic d the density $\sqrt{-g}$ is to be read as $\sqrt{|g|}$. Index symmetrization is *not* normalized: $h^{(\mu\nu)} \equiv h^{\mu\nu} + h^{\nu\mu}$. Fourier transforms are $\tilde{\phi}^{\mu\nu}(p) = \int d^d x e^{ip \cdot x} \phi^{\mu\nu}(x)$, and likewise for $\tilde{G}^\lambda_{\mu\nu}$ and for the gauge parameter $\tilde{\omega}^\mu$, so that $\phi^{\mu\nu}(x) = \int_p e^{-ip \cdot x} \tilde{\phi}^{\mu\nu}(p)$ with $\int_p \equiv \int d^d p / (2\pi)^d$; a derivative acting on a field of momentum p becomes $-ip_\mu$, and a product of fields becomes a convolution. Table I lists the objects used throughout.

b. Compact notation. In the main text we suppress indices. A field carries its momentum as a subscript: ϕ_p stands for the whole tensor $\tilde{\phi}^{\mu\nu}(p)$, G_p for $\tilde{G}^\lambda_{\mu\nu}(p)$, ω_p for $\tilde{\omega}^\mu(p)$; the index structure of each is that of Table I. The objects that act on fields are of three kinds.

- A *kernel* K is a multilinear form on fields: $K(A, B)$ or $K(A, B, C)$ denotes the full contraction of the indices of each argument with the corresponding index group of the kernel K . For instance, $K^{\phi G}(\phi_{-p}, G_p) \equiv \tilde{\phi}^{\mu\nu}(-p) K^{\phi G}_{\mu\nu\lambda}{}^{\rho\sigma}(p) \tilde{G}^\lambda_{\rho\sigma}(p)$.
- A *linear map* $D(p)$ takes the gauge parameter to a field: $D_{\phi^0}(p) \omega_p \equiv D_{\phi^0}{}^{\mu\nu}{}_\alpha(p) \tilde{\omega}^\alpha(p)$ is a symmetric tensor, and likewise $D_{G^0}(p) \omega_p$. The label of a D names the field it produces and the order in κ of the term of the transformation from which it arises: D_{ϕ^0} is the field-independent part of $\delta\phi$, D_{ϕ^1} the part linear in the fields, Eq. (2.5) below.
- A *bilinear map* $D(p, q)$ takes a field and the gauge parameter to a field,

$$D_{\phi^1}(p, q)(\phi_{p-q}, \omega_q) \equiv \tilde{\phi}^{\alpha\beta}(p-q) \tilde{\omega}^\rho(q) D_{\phi^1}{}^{\mu\nu}{}_{\alpha\beta\rho}(p, q),$$

and likewise $D_{G^1}(p, q)(G_{p-q}, \omega_q)$.

The index structure of every object is given in Table II and in the appendices; the compact form is only a shorthand for those expressions.

Symbol	Meaning	Indices	Momenta	Dimension
κ	gravitational coupling			$(2-d)/2$
$\phi^{\mu\nu}; \phi_p$	density perturbation, Eq. (2.1)	$\mu\nu$ symmetric	one	$(d-2)/2; -(d+2)/2$
$G^\lambda{}_{\mu\nu}; G_p$	trace-adjusted connection, Eq. (2.1)	$\mu\nu$ symmetric	one	$d/2; -d/2$
$\omega^\mu; \omega_p$	gauge parameter	vector	one	$(d-4)/2; -(d+4)/2$
$D_{\phi^0}(p), D_{G^0}(p)$	$\delta^{(0)}\phi, \delta^{(0)}G$ as maps on ω	App. B	p	1, 2
$D_{\phi^1}(p, q), D_{G^1}(p, q)$	$\delta^{(1)}\phi, \delta^{(1)}G$ as maps on (field, ω)	App. B	p, q	1, 1
$K^{\phi\phi}, K^{\phi G}, K^{GG}$	quadratic kernels of S_2	Table II	p	2, 1, 0
$K^{\phi GG}$	cubic kernel of S_3	Table II	none	0
$\mathcal{W}^\phi, \mathcal{W}^G, \mathcal{W}^{31}, \dots, \mathcal{W}^{34}$	the six Noether identities	App. D	$p; p_i$	

TABLE I. Notation and mass dimensions. For the fields and the gauge parameter the two entries in the last column refer to configuration and momentum space; $[\int_p] = d$.

c. Fields. The independent fields are the perturbation of the Goldberg density [7] and the trace-adjusted connection, both normalized with the gravitational coupling κ ,

$$\kappa \phi^{\mu\nu} = \sqrt{-g} g^{\mu\nu} - \eta^{\mu\nu}, \quad \kappa G^\lambda{}_{\mu\nu} = \Gamma^\lambda{}_{\mu\nu} - \frac{1}{2}(\delta^\lambda{}_\mu \Gamma^\sigma{}_{\nu\sigma} + \delta^\lambda{}_\nu \Gamma^\sigma{}_{\mu\sigma}), \quad (2.1)$$

both real and symmetric in $\mu\nu$. Here $\Gamma^\lambda{}_{\mu\nu}$ is an independent symmetric affine connection, the Palatini variable [8, 9], and not the Levi-Civita connection built from $g_{\mu\nu}$: for the action found below, the field equation of Γ identifies the two when $d > 2$ and the metric is nondegenerate, and that equation is not imposed in the construction. The second relation in Eq. (2.1) is a linear change of variables, invertible for $d \neq 1$. Taking its trace gives $\Gamma^\sigma{}_{\nu\sigma} = \frac{2\kappa}{1-d} g_\nu$, with $g_\nu \equiv G^\sigma{}_{\nu\sigma}$, and substituting back,

$$\Gamma^\lambda{}_{\mu\nu} = \kappa \left[G^\lambda{}_{\mu\nu} + \frac{1}{1-d} (\delta^\lambda{}_\mu g_\nu + \delta^\lambda{}_\nu g_\mu) \right], \quad (2.2)$$

Eqs. (6)–(8) of Ref. [13]. The trace adjustment is what cancels the $\partial_\nu \Gamma^\lambda{}_{\mu\lambda}$ term of $h^{\mu\nu} R_{\mu\nu}(\Gamma)$ against the trace part of $\partial_\lambda \Gamma^\lambda{}_{\mu\nu}$, leaving the single derivative term of Eq. (5.1) below. With $[\kappa] = (2-d)/2$ in mass units, $\kappa\phi$ is dimensionless, $[\phi] = (d-2)/2$, and $[G] = d/2$; in $d = 4$, $[\phi] = 1$ and $[G] = 2$. Note that G does not have the dimension of a connection: it is the trace-adjusted connection divided by κ , in the same way that ϕ is the metric perturbation divided by κ . The definitions in Eqs. (2.1)–(2.2) serve only to fix the transformation laws below; from then on ϕ and G are treated as independent fields, and no relation between G and a metric is assumed.

d. Transformations. Under an infinitesimal diffeomorphism $x^\mu \rightarrow x^\mu - \kappa \omega^\mu(x)$ the fields transform as

$$\delta\phi^{\mu\nu} = \underbrace{\eta^{\mu\nu}\partial\cdot\omega - \partial^{(\mu}\omega^{\nu)}}_{\delta^{(0)}\phi^{\mu\nu}} + \kappa \underbrace{[\partial_\rho(\phi^{\mu\nu}\omega^\rho) - \partial_\rho\omega^{(\mu}\phi^{\nu)\rho}]}_{\delta^{(1)}\phi^{\mu\nu}}, \quad (2.3)$$

$$\delta G^\lambda_{\mu\nu} = \underbrace{\partial_\mu\partial_\nu\omega^\lambda - \frac{1}{2}\delta^\lambda_{(\mu}\partial_{\nu)}\partial\cdot\omega}_{\delta^{(0)}G^\lambda_{\mu\nu}} + \kappa \underbrace{[\omega^\rho\partial_\rho G^\lambda_{\mu\nu} - G^\rho_{\mu\nu}\partial_\rho\omega^\lambda + G^\lambda_{\rho(\mu}\partial_{\nu)}\omega^\rho]}_{\delta^{(1)}G^\lambda_{\mu\nu}}. \quad (2.4)$$

The gauge parameter has $[\omega] = (d-4)/2$, so that it is dimensionless in $d=4$. The split into a field-independent contribution $\delta^{(0)}$ and a field-linear contribution $\kappa\delta^{(1)}$, of order κ , is what organizes the whole construction. Eq. (2.3) follows from the transformation of $h^{\mu\nu} = \eta^{\mu\nu} + \kappa\phi^{\mu\nu}$ as a tensor density of weight one, with the background $\eta^{\mu\nu}$ held fixed; Eq. (2.4) follows from the transformation of an affine connection through Eq. (2.1). Both are derived in Appendix A; they are Eqs. (17b) and (18) of Ref. [13]. These two laws are the only input of the construction beyond the class of actions defined in the next section.

In momentum space the derivatives become momenta and $\delta^{(1)}$ becomes a convolution:

$$\delta\phi_p = D_{\phi^0}(p)\omega_p + \kappa \int_q D_{\phi^1}(p, q)(\phi_{p-q}, \omega_q), \quad \delta G_p = D_{G^0}(p)\omega_p + \kappa \int_q D_{G^1}(p, q)(G_{p-q}, \omega_q). \quad (2.5)$$

This form makes the structure explicit: $\delta^{(0)}$ is a linear map on the gauge parameter, $\delta^{(1)}$ a bilinear map on the pair (field, gauge parameter). The four tensors $D_{\phi^0}, D_{\phi^1}, D_{G^0}, D_{G^1}$ are listed in Appendix B.

III. THE ACTION ANSATZ

We look for an action $S = S_2 + \kappa S_3$, quadratic plus cubic in the fields, of the form

$$\begin{aligned} S_2 &= \int_p \left[K^{\phi\phi}(\phi_{-p}, \phi_p) + K^{\phi G}(\phi_{-p}, G_p) + K^{GG}(G_{-p}, G_p) \right], \\ S_3 &= \int_p \int_q K^{\phi GG}(\phi_{p+q}, G_{-p}, G_{-q}). \end{aligned} \quad (3.1)$$

The kernels $K^{\phi\phi}$ and $K^{\phi G}$ depend on the momentum p ; K^{GG} and $K^{\phi GG}$ do not, for the reason given below. The quadratic part is the most general translation-invariant local expression within the coupling-power assignment and the parity-even tensor class specified below. The cubic part is restricted to the single structure ϕGG ; this is an assumption, discussed below.

a. Derivative counting. The dimensions of the fields are fixed by the transformation laws: $\kappa\omega$ is a coordinate displacement, so $[\kappa\omega] = -1$, and Eqs. (2.3)–(2.4) then give $[\phi] = (d-2)/2$ and $[G] = d/2$, consistent with Eq. (2.1) (see Table I). With these assignments S carries no prefactor, and the ansatz prescribes the power of the coupling at each order in the fields, κ^{n-2} in S_n . A local monomial with a fields ϕ , b fields G and r derivatives, multiplied by κ^{a+b-2} , then has mass dimension $d+r+b-2$, so that a dimensionless action requires $r+b=2$. This gives two derivatives for $K^{\phi\phi}$, one for $K^{\phi G}$ and none for K^{GG} ; at cubic order it allows $\phi\phi\phi$ with two derivatives, $\phi\phi G$ with one and ϕGG with none, and excludes GGG , which would require $r=-1$. The derivative bounds thus rest on locality, dimensional consistency and the coupling-power assignment; they do not by themselves bound the number of fields ϕ . The cubic vertex $K^{\phi GG}$ is purely algebraic, which is the feature of the first-order formulation that makes the determination finite.

b. The cubic sector. The restriction of S_3 to the structure ϕGG is an assumption: the counting above also allows $\phi\phi G$ with one derivative and $\phi\phi\phi$ with two, and both would keep the field equation of G algebraic. The restriction is motivated by the polynomial structure of the Goldberg–Palatini action, in which the connection appears at most quadratically and the only derivative acts in the ϕG term; it defines the class in which we investigate whether the Noether identities admit a completion terminating at cubic order. The uniqueness result of Section V is conditional on this restriction, and its relaxation is discussed in Section VII.

c. Tensor bases. Each kernel is expanded in a basis of tensors built from $\eta^{\mu\nu}$ and p^μ , with the index symmetries inherited from the fields it multiplies; parity-odd structures involving the Levi-Civita tensor are not considered. The bases are generated by symmetrizing, under the index symmetries of each kernel, the tensor monomials, that is, the products of η ’s and p ’s with the required indices and degree in p ; their linear independence is verified through Gram determinants (Appendix C). Table II gives the outcome. For $d \geq 4$ the four bases are independent and contain $4 + 6 + 5 + 16 = 31$ unknown coefficients in total.

IV. NOETHER IDENTITIES AND THEIR DECOMPOSITION

Invariance of S under Eqs. (2.3)–(2.4), ordered by powers of κ , or equivalently by the number of fields, gives three identities,

$$\delta^{(0)}S_2 = 0, \quad \delta^{(0)}S_3 + \delta^{(1)}S_2 = 0, \quad \delta^{(1)}S_3 = 0, \quad (4.1)$$

Kernel	Index symmetries	Momentum	Monomials	Basis dimension
$K_{\mu\nu\rho\sigma}^{\phi\phi}$	$\mu \leftrightarrow \nu, \rho \leftrightarrow \sigma, (\mu\nu) \leftrightarrow (\rho\sigma)$	p^2	$p^2\eta\eta, pp\eta$	4
$K_{\mu\nu\lambda\rho\sigma}^{\phi G}$	$\mu \leftrightarrow \nu, \rho \leftrightarrow \sigma$	p	$p\eta\eta$	6
$K_{\lambda\mu\nu\rho\sigma\tau}^{GG}$	$\mu \leftrightarrow \nu, \sigma \leftrightarrow \tau, (\lambda\mu\nu) \leftrightarrow (\rho\sigma\tau)$	p^0	$\eta\eta\eta$	5
$K_{\mu\nu\lambda\alpha\beta\rho\sigma\tau}^{\phi GG}$	$\mu \leftrightarrow \nu, \alpha \leftrightarrow \beta, \sigma \leftrightarrow \tau, (\lambda\alpha\beta) \leftrightarrow (\rho\sigma\tau)$	p^0	$\eta\eta\eta\eta$	16

TABLE II. The four kernels of the ansatz (3.1), their symmetries, and the dimension of the tensor basis in which each is expanded, for $d \geq 4$.

where $\delta^{(m)}S_n \equiv \int (\delta S_n / \delta A) \delta^{(m)}A$ summed over the fields $A = \phi, G$. Each identity holds for arbitrary ω and arbitrary field configurations, so it can be decomposed further according to which fields appear. Taking functional derivatives with respect to the fields and setting the fields to zero turns each piece into an algebraic tensor identity among the kernels, in momentum space. The decomposition is summarized in Table III; the six identities are written out in Appendix D. The resulting relations among the action kernels are also called classical Ward identities, the terminology used in the accompanying notebooks; we keep “Noether identities” to emphasize their derivation from Noether’s second theorem. They concern the classical action before gauge fixing, not the Green functions or the BRST identities of a gauge-fixed theory.

Identity	From	Field sector	Kernels involved	Projection basis
$\mathcal{W}^\phi$	$\delta^{(0)}S_2$	ϕ	$K^{\phi\phi}, K^{\phi G}$	3
$\mathcal{W}^G$	$\delta^{(0)}S_2$	G	$K^{\phi G}, K^{GG}$	6
$\mathcal{W}^{31}$	$\delta^{(1)}S_2$	$\phi\phi$	$K^{\phi\phi}$	7
$\mathcal{W}^{32}$	$\delta^{(0)}S_3 + \delta^{(1)}S_2$	ϕG	$K^{\phi GG}, K^{\phi G}$	23
$\mathcal{W}^{33}$	$\delta^{(0)}S_3 + \delta^{(1)}S_2$	GG	$K^{\phi GG}, K^{GG}$	20
$\mathcal{W}^{34}$	$\delta^{(1)}S_3$	ϕGG	$K^{\phi GG}$	95

TABLE III. Decomposition of the Noether identities (4.1) into field sectors. The last column is the generic-dimension size of the projection basis onto which each identity is projected to extract scalar equations for the coefficients, with the momentum specialization described in the text; dimension-specific dependencies, such as that of the 95-element basis at $d = 4$, are discussed in Appendix D.

Two remarks on the structure of Table III contain most of the argument. First, $\mathcal{W}^{31}$ involves the $\phi\phi$ kernel alone, because S_3 contains two factors of G and $\delta^{(1)}G$ is linear in G : nothing in the cubic sector can contribute to the $\phi\phi$ sector. Second, the cubic kernel $K^{\phi GG}$ enters $\mathcal{W}^{32}$ and $\mathcal{W}^{33}$, which relate it to the quadratic kernels through the field-dependent part $\delta^{(1)}$ of the transformations, and $\mathcal{W}^{34}$, which constrains the cubic kernel alone. Within the chosen ansatz, these identities couple the cubic vertex to the quadratic action; the solution below establishes that no independent cubic coefficient remains.

Each identity is turned into scalar equations by projecting it onto a basis of tensors with the same index structure and momentum dependence; since the identity vanishes, so does every projection. Conversely, for an identity in the span of a projection basis with nondegenerate Gram matrix, the vanishing of all projections is equivalent to the vanishing of the identity; the one exception, $\mathcal{W}^{34}$ at $d = 4$, is discussed in Appendix D. In the identities with two or three momenta the momenta are identified, $p_1 = p_2 = p_3 \equiv p$, so that every projection basis is built from η and a single vector; the dimensions in Table III refer to these bases. Since an identity that holds for arbitrary momenta holds in particular for equal ones, the specialization can only weaken the system: it yields necessary conditions, never spurious ones, and the solution space of the specialized system contains that of the full one. The specialized system already has a one-dimensional solution space (Section V), and the Goldberg–Palatini action, which is invariant for arbitrary momenta, belongs to it; the two solution spaces therefore coincide and nothing is lost. The identification that establishes invariance for arbitrary momenta is discussed in Section VI.

V. SOLUTION

The resulting linear system is solved in a sequence that follows the structure of Table III, and each step has a direct interpretation. The sequence is shown in Table IV.

Step 1 shows that the identities force the purely metric kernel to vanish, $K^{\phi\phi} = 0$, before any comparison with a known action: in this formalism the dynamics of ϕ is carried entirely by its coupling to G . After $K^{\phi\phi} = 0$ is imposed, the quadratic identities of Step 2 leave two parameters in the quadratic action, the overall constant and one relative coefficient. Step 3 removes the remaining relative freedom: the cubic identity in the ϕG sector fixes that coefficient and expresses all sixteen coefficients of $K^{\phi GG}$ in terms of the overall constant.

Step	Identity	Constraints imposed	Remaining freedom
1	$\mathcal{W}^{31}$	all 4 of $K^{\phi\phi}$, to zero	27
2	$\mathcal{W}^\phi, \mathcal{W}^G$	9 of the 11 in $K^{\phi G}, K^{GG}$	2 in S_2 ; 16 in $K^{\phi GG}$
3	$\mathcal{W}^{32}$	the remaining relative freedom in S_2 ; all 16 of $K^{\phi GG}$	1 (overall constant)
4	$\mathcal{W}^{33}, \mathcal{W}^{34}$	none: both are satisfied identically	1

TABLE IV. Sequential solution of the constraint system. The third column counts independent constraints, that is, parameters eliminated at each step; which individual coefficients are expressed in terms of which depends on the parametrization chosen.

Step 4 confirms that the two remaining identities impose no further condition. The solution is therefore unique up to an overall constant.

Transforming back to configuration space, and writing $g_\mu \equiv G^\sigma{}_{\mu\sigma}$ for the trace of G and $\text{tr}(G_\mu G_\nu) \equiv G^\lambda{}_{\mu\sigma} G^\sigma{}_{\nu\lambda}$ for the matrix trace over the upper and one lower index, the action selected by the identities reads

$$S = \int d^d x \left[-G^\lambda{}_{\mu\nu} \partial_\lambda \phi^{\mu\nu} + (\eta + \kappa \phi)^{\mu\nu} \left(-\text{tr}(G_\mu G_\nu) + \frac{1}{d-1} g_\mu g_\nu \right) \right], \quad (5.1)$$

and multiplying it by $\frac{1}{2}$ gives, modulo a boundary term, the Goldberg–Palatini action $\frac{1}{2\kappa^2} \int d^d x h^{\mu\nu} R_{\mu\nu}(\Gamma)$ written in the variables (2.1), with $h^{\mu\nu} = \eta^{\mu\nu} + \kappa \phi^{\mu\nu}$; Ref. [10] writes the same action with $\kappa = 1$, Eqs. (3.4)–(3.5) there. With the field dimensions and coupling-power assignment adopted here, the overall constant left undetermined by the identities is dimensionless. The explicit momentum-space kernels are given in Appendix E.

VI. CHECKS AND SCOPE

a. Generic momenta. The solution of Section V was obtained with the momenta of the cubic identities identified, and the argument of Section IV shows that nothing is lost provided the resulting action is indeed invariant for arbitrary momenta. That it is follows from its identification with $h^{\mu\nu} R_{\mu\nu}(\Gamma)$ in Section V, a scalar density under the transformations (2.3)–(2.4) by construction: its variation is a total derivative, so the action is invariant for gauge parameters of compact support, or for boundary conditions that remove the boundary contribution; this closes the argument. The specialization enters only the cubic identities $\mathcal{W}^{31}$ – $\mathcal{W}^{34}$; $\mathcal{W}^\phi$ and $\mathcal{W}^G$ involve a single momentum and are used as they stand.

b. Reduction to the metric formulation. Writing the GG term of Eq. (5.1) as $\frac{1}{2}M(\eta + \kappa\phi)(G, G)$, with $M(h)$ the bilinear form $M(h)(G, G) = 2h^{\mu\nu}(-\text{tr}(G_\mu G_\nu) + \frac{1}{d-1}g_\mu g_\nu)$, linear in h , the field equation of G is algebraic, $M(\eta + \kappa\phi)G = \partial\phi$; for $d > 2$ and nondegenerate h the form $M(h)$ is invertible, and the solution $G = M^{-1}(\eta + \kappa\phi)\partial\phi$ substituted back gives

$$S = -\frac{1}{2} \int d^d x \, M^{-1}(\eta + \kappa\phi)(\partial\phi, \partial\phi), \quad (6.1)$$

the second-order Einstein–Hilbert action in the Goldberg variable, Eq. (3.13) of Ref. [10], to all orders in ϕ , modulo a boundary term and with the normalization of Section V. Since M is linear in its argument, the inverse has the formal perturbative expansion

$$M^{-1}(\eta + \kappa\phi) = M^{-1}(\eta) - \kappa M^{-1}(\eta)M(\phi)M^{-1}(\eta) + \dots, \quad (6.2)$$

Eq. (3.18) of the same reference: inserted in Eq. (6.1), the first term gives the quadratic graviton action and each further term an n -point vertex. In particular, with $G_0 \equiv M^{-1}(\eta)\partial\phi$ the solution of the field equation of G at lowest order, the second term of Eq. (6.2) gives the cubic term

$$S^{(3)} = \frac{\kappa}{2} \int d^d x \, M(\phi)(G_0, G_0), \quad (6.3)$$

the cubic interaction in closed form. The infinite tower of self-interactions of the metric formulation is thus fixed by the three kernels of the first-order action.

The reduction has been checked symbolically at quadratic and cubic order against the Lagrangians of Capper, Leibbrandt and Ramón Medrano (CLM) [14], Eqs. (2.10) and (2.16) of that reference, which are written in the same metric-density perturbation $\sqrt{-g}g^{\mu\nu} = \eta^{\mu\nu} + \kappa\phi^{\mu\nu}$. Starting from the bootstrap kernels, the accompanying xAct notebook (Appendix F) derives M_0^{-1} , with $M_0 \equiv M(\eta)$, verifies the left and right inverse relations, constructs $G_0 = M_0^{-1}\partial\phi$, and evaluates the quadratic Lagrangian from Eq. (6.1) and the cubic one from Eq. (6.3). The conventions must be kept explicit: CLM write $2K^{-2} \int d^d x \, h^{\mu\nu} R_{\mu\nu}^{\text{CLM}}$ with $R_{\mu\nu}^{\text{CLM}} = -R_{\mu\nu}$ in our convention, whereas Eq. (5.1) corresponds to $\kappa^{-2} \int d^d x \, h^{\mu\nu} R_{\mu\nu}$ modulo a boundary term. With $K = \kappa$ and $\mathcal{L}_{(j)}^{\text{CLM}}$ the coefficients of their expansion $\mathcal{L} = \sum_{j \geq 2} K^{j-2} \mathcal{L}_{(j)}^{\text{CLM}}$, the symbolic checks give

$$\mathcal{L}^{(2)} + \frac{1}{2} \mathcal{L}_{(2)}^{\text{CLM}} = \partial_a J^a, \quad J^a = \frac{1}{2}(\phi^{bc} \partial_c \phi^a{}_b - \phi^{ab} \partial_c \phi^c{}_b), \quad (6.4)$$

$$\mathcal{L}^{(3)} + \frac{\kappa}{2} \mathcal{L}_{(3)}^{\text{CLM}} = 0, \quad (6.5)$$

with indices raised and lowered with η .¹ The cubic equality is local and requires neither integration by parts nor the field equations of ϕ ; the quadratic actions agree under boundary conditions that remove the integral of $\partial_a J^a$. The corresponding three-graviton vertices therefore agree once the action, Fourier and symmetrization conventions are matched, without a separate transcription of CLM's momentum-space vertex.

c. Scope. The inputs of the construction are the field content, the transformation laws (2.3)–(2.4), the bounds on derivatives, and the restriction of the cubic sector to ϕGG . The result is uniqueness within that class. It does not establish uniqueness in a class containing further powers of the fields, additional derivatives, or independent dimensionful couplings. The overall constant is not fixed by the homogeneous invariance conditions; it requires an external normalization, such as matching the weak-field equations to the Newtonian limit with a specified normalization of the matter source, which for a fixed convention for κ determines it. The construction is classical throughout; whether the first- and second-order formulations remain equivalent beyond tree level, a question studied in Ref. [10], is not addressed here.

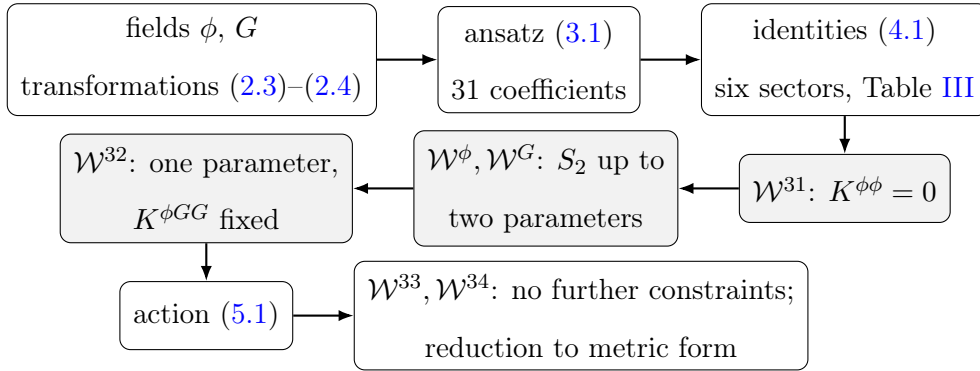


FIG. 1. The determination of the action. Inputs on the top row; the sequential solution of Table IV on the middle row; output and checks on the bottom row.

VII. CONCLUSIONS

The procedure described here identifies the Goldberg–Palatini action through the Noether identities of its prescribed field transformations. Within the local, Lorentz-covariant, parity-

¹ In transcribing Eq. (2.16) of Ref. [14], the first term of its second group is read as $\phi_{\mu\tau}\phi_{\mu\sigma,\lambda}\phi_{\lambda\tau,\sigma}$, which restores consistent index contractions in the printed expression.

even ansatz with the prescribed derivative orders and the cubic sector restricted to ϕGG , the 31 coefficients of the kernel bases (for $d \geq 4$) are reduced to a single overall constant. In particular, the identities eliminate the quadratic $\phi\phi$ kernel and fix the relative coefficients of the mixed ϕG term, the algebraic GG term, and the cubic ϕGG interaction. The resulting structure is that of first-order Einstein gravity.

Individual Noether identities can leave components of a kernel undetermined, since they constrain tensor contractions. Uniqueness here follows from imposing all identities simultaneously within the chosen ansatz: the field-dependent transformations link the quadratic and cubic sectors and remove the freedom left by the quadratic identities alone. The resulting homogeneous linear system has a one-dimensional solution space, corresponding to the overall normalization of the action. Any further independent solution would define another invariant action in the same class. Thus uniqueness is a property of this restricted system, not a general consequence of Noether identities. The connection is not a field exempt from the procedure: its variation enters every sector of the identity except the $\phi\phi$ one, and the coupled identities determine the kernels involving it; once the preceding constraints are imposed, the cubic vertex is fixed by the sector that couples the two fields, $\mathcal{W}^{32}$. The solution was obtained with the momenta of the cubic identities identified; its invariance for arbitrary momenta follows from its identification with the Goldberg–Palatini action.

For $d > 2$ and nondegenerate metric density the algebraic equation for G has a unique solution, so fixing the first-order action fixes the second-order one completely. Eliminating G yields the metric action as the quadratic form (6.1), modulo a boundary term and with the overall normalization inherited from the first-order action, and the formal expansion of its kernel in powers of κ produces every graviton self-interaction vertex of the second-order theory. A construction that closes at cubic order in the first-order variables therefore determines an infinite set of vertices in the metric variables; this is the practical content of the classical equivalence between the two formulations, and it is what makes the first-order variables the natural setting for the determination.

The method makes the role of symmetry explicit at each order in the fields. It avoids selecting an energy–momentum tensor as the source of successive self-couplings and instead imposes off-shell relations directly on the action kernels. Its practical advantage is that the polynomial first-order formulation turns the determination of the action into a finite tensor-algebra problem.

A natural extension is to enlarge the interaction ansatz while retaining explicit control over locality and derivative order. Terms involving additional powers of ϕ , or derivatives compatible with other cubic field combinations, would test whether the restriction to ϕGG can be relaxed without introducing new invariant structures. For Yang–Mills theory the tree-amplitude version of the argument is known to close and to fix the interaction structure under its stated assumptions [6]; the action-level version with tensor bases would provide a direct check of the method, while higher-derivative gravitational actions would allow it to probe a broader space of gauge-invariant dynamics.

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Appendix A: Derivation of the transformation laws

This appendix derives Eqs. (2.3) and (2.4) from the transformation properties of $g_{\mu\nu}$ and $\Gamma^\lambda_{\mu\nu}$ under coordinate changes. They are Eqs. (17b) and (18) of Ref. [13], whose Eq. (17a) is the transformation of $h^{\mu\nu}$ itself, Eq. (A3) below; the parameter θ of that reference is $-\kappa\omega$ here, and that reference sets $\kappa = 1$.

a. Conventions. Under the infinitesimal coordinate change $x'^\mu = x^\mu - \varepsilon^\mu(x)$ we define the variation of a field as $\delta T(x) \equiv T'(x) - T(x)$. For a tensor this is the Lie derivative along ε , $\delta T = \mathcal{L}_\varepsilon T$; for the objects needed below,

$$\mathcal{L}_\varepsilon V^\mu = \varepsilon^\rho \partial_\rho V^\mu - V^\rho \partial_\rho \varepsilon^\mu, \quad \mathcal{L}_\varepsilon T_\nu = \varepsilon^\rho \partial_\rho T_\nu + T_\rho \partial_\nu \varepsilon^\rho, \quad (\text{A1})$$

$$\mathcal{L}_\varepsilon X^\lambda_{\mu\nu} = \varepsilon^\rho \partial_\rho X^\lambda_{\mu\nu} - X^\rho_{\mu\nu} \partial_\rho \varepsilon^\lambda + X^\lambda_{\rho\nu} \partial_\mu \varepsilon^\rho + X^\lambda_{\mu\rho} \partial_\nu \varepsilon^\rho. \quad (\text{A2})$$

$\mathcal{L}_\varepsilon$ is linear and obeys the Leibniz rule, and it annihilates the Kronecker delta: $\mathcal{L}_\varepsilon \delta^\lambda_\mu = -\delta^\rho_\mu \partial_\rho \varepsilon^\lambda + \delta^\lambda_\rho \partial_\mu \varepsilon^\rho = 0$. At the end we set $\varepsilon = \kappa\omega$, which is the parametrization of Section II.

b. The metric density. From $\delta g_{\mu\nu} = \mathcal{L}_\varepsilon g_{\mu\nu}$ and $\delta g^{\mu\nu} = -g^{\mu\alpha} g^{\nu\beta} \delta g_{\alpha\beta}$ one has $\delta g^{\mu\nu} = \varepsilon^\rho \partial_\rho g^{\mu\nu} - g^{\rho\nu} \partial_\rho \varepsilon^\mu - g^{\mu\rho} \partial_\rho \varepsilon^\nu$, and from $\delta \sqrt{-g} = \frac{1}{2} \sqrt{-g} g^{\alpha\beta} \delta g_{\alpha\beta}$ together with $\partial_\rho \sqrt{-g} = \frac{1}{2} \sqrt{-g} g^{\alpha\beta} \partial_\rho g_{\alpha\beta}$, $\delta \sqrt{-g} = \varepsilon^\rho \partial_\rho \sqrt{-g} + \sqrt{-g} \partial_\rho \varepsilon^\rho$. The product rule then gives, for $h^{\mu\nu} \equiv \sqrt{-g} g^{\mu\nu}$,

$$\delta h^{\mu\nu} = \partial_\rho (h^{\mu\nu} \varepsilon^\rho) - h^{\rho\nu} \partial_\rho \varepsilon^\mu - h^{\mu\rho} \partial_\rho \varepsilon^\nu, \quad (\text{A3})$$

the transformation of a contravariant tensor density of weight one. Holding the background $\eta^{\mu\nu}$ fixed, so that $\delta h^{\mu\nu} = \kappa \delta \phi^{\mu\nu}$, we insert $h^{\mu\nu} = \eta^{\mu\nu} + \kappa \phi^{\mu\nu}$ and $\varepsilon = \kappa \omega$ and divide by κ ,

$$\delta \phi^{\mu\nu} = \eta^{\mu\nu} \partial \cdot \omega - \partial^{(\mu} \omega^{\nu)} + \kappa \left[\partial_\rho (\phi^{\mu\nu} \omega^\rho) - \partial_\rho \omega^{(\mu} \phi^{\nu)\rho} \right], \quad (\text{A4})$$

which is Eq. (2.3).

c. The connection. An affine connection transforms so that $\nabla_\mu V^\nu$ is a tensor for every V ; the infinitesimal form of the resulting law is the Lie derivative of a $(1, 2)$ tensor plus an inhomogeneous term,

$$\delta \Gamma^\lambda_{\mu\nu} = \mathcal{L}_\varepsilon \Gamma^\lambda_{\mu\nu} + \partial_\mu \partial_\nu \varepsilon^\lambda. \quad (\text{A5})$$

Here and below, when applied to the connection, to its trace or to the trace-adjusted connection, $\mathcal{L}_\varepsilon$ denotes only the corresponding tensor-form expression; the inhomogeneous terms, which make these objects non-tensorial, are displayed separately.

d. The trace. Let $T_\nu \equiv \Gamma^\lambda_{\nu\lambda}$. Contracting λ with μ in Eq. (A5), the second and third terms of the Lie derivative (A2) become $-\Gamma^\rho_{\lambda\nu} \partial_\rho \varepsilon^\lambda + \Gamma^\lambda_{\rho\nu} \partial_\lambda \varepsilon^\rho$ and cancel after renaming the dummy indices, while the inhomogeneous term becomes $\partial_\nu (\partial \cdot \varepsilon)$:

$$\delta T_\nu = \mathcal{L}_\varepsilon T_\nu + \partial_\nu (\partial \cdot \varepsilon). \quad (\text{A6})$$

e. The trace-adjusted connection. By Eq. (2.1), $\Gamma^\lambda_{\mu\nu} = \kappa G^\lambda_{\mu\nu} + \frac{1}{2} (\delta^\lambda_\mu T_\nu + \delta^\lambda_\nu T_\mu)$, and therefore $\delta(\kappa G) = \delta\Gamma - \frac{1}{2} (\delta^\lambda_\mu \delta T_\nu + \delta^\lambda_\nu \delta T_\mu)$. In $\delta\Gamma$, Eq. (A5), the Lie derivative acts linearly on the two pieces of Γ , and since it annihilates δ^λ_μ , $\mathcal{L}_\varepsilon (\delta^\lambda_\mu T_\nu) = \delta^\lambda_\mu \mathcal{L}_\varepsilon T_\nu$. Using Eq. (A6) for δT , the terms $\frac{1}{2} \delta^\lambda_\mu \mathcal{L}_\varepsilon T_\nu + \frac{1}{2} \delta^\lambda_\nu \mathcal{L}_\varepsilon T_\mu$ cancel between $\delta\Gamma$ and the trace correction, and what survives of the latter is the inhomogeneous part of Eq. (A6):

$$\delta(\kappa G^\lambda_{\mu\nu}) = \kappa \mathcal{L}_\varepsilon G^\lambda_{\mu\nu} + \partial_\mu \partial_\nu \varepsilon^\lambda - \frac{1}{2} (\delta^\lambda_\mu \partial_\nu + \delta^\lambda_\nu \partial_\mu) \partial \cdot \varepsilon. \quad (\text{A7})$$

With $\varepsilon = \kappa \omega$ and Eq. (A2) for $\mathcal{L}_\omega G$, this is Eq. (2.4). The trace adjustment thus leaves the two-derivative term $\partial_\mu \partial_\nu \omega^\lambda$ unchanged and subtracts $\frac{1}{2} (\delta^\lambda_\mu \partial_\nu + \delta^\lambda_\nu \partial_\mu) \partial \cdot \omega$, so that the trace

of the variation is $\delta^{(0)}G^\lambda{}_{\nu\lambda} = \frac{1-d}{2}\partial_\nu\partial\cdot\omega$, consistent with $\Gamma^\sigma{}_{\nu\sigma} = \frac{2\kappa}{1-d}g_\nu$; this is why $\delta^{(0)}G$ is not simpler than $\delta^{(0)}\Gamma$ even though the action in the variable G is. Both Eq. (A6) and the final result were verified symbolically (Appendix F).

Appendix B: Transformations in momentum space

With the Fourier conventions of Section II, Eq. (2.5) reads, with all indices displayed,

$$\delta\tilde{\phi}^{\mu\nu}(p) = D_{\phi^0}{}^{\mu\nu}{}_\alpha(p)\tilde{\omega}^\alpha(p) + \kappa \int_q \tilde{\phi}^{\alpha\beta}(p-q)\tilde{\omega}^\rho(q)D_{\phi^1}{}^{\mu\nu}{}_{\alpha\beta\rho}(p,q), \quad (\text{B1})$$

$$\delta\tilde{G}^\lambda{}_{\mu\nu}(p) = D_{G^0}{}^\lambda{}_{\mu\nu\alpha}(p)\tilde{\omega}^\alpha(p) + \kappa \int_q \tilde{G}^\alpha{}_{\beta\gamma}(p-q)\tilde{\omega}^\rho(q)D_{G^1}{}^\lambda{}_{\mu\nu\alpha}{}^{\beta\gamma}{}_\rho(p,q), \quad (\text{B2})$$

with

$$D_{\phi^0}{}^{\mu\nu}{}_\alpha(p) = -i(\eta^{\mu\nu}p_\alpha - p^{(\mu}\delta_\alpha^{\nu)}), \quad (\text{B3})$$

$$D_{\phi^1}{}^{\mu\nu}{}_{\alpha\beta\rho}(p,q) = -i(\delta_\alpha^\mu\delta_\beta^\nu p_\rho - q_\beta\delta_\alpha^{(\mu}\delta_\rho^{\nu)}), \quad (\text{B4})$$

$$D_{G^0}{}^\lambda{}_{\mu\nu\alpha}(p) = -p_\mu p_\nu \delta_\alpha^\lambda + \frac{1}{2}\delta_{(\mu}^\lambda p_{\nu)}p_\alpha, \quad (\text{B5})$$

$$D_{G^1}{}^\lambda{}_{\mu\nu\alpha}{}^{\beta\gamma}{}_\rho(p,q) = -i[\delta_\alpha^\lambda\delta_\mu^\beta\delta_\nu^\gamma(p-q)_\rho - q_\alpha\delta_\mu^\beta\delta_\nu^\gamma\delta_\rho^\lambda + \delta_\alpha^\lambda\delta_\rho^\beta\delta_\mu^\gamma q_\nu]. \quad (\text{B6})$$

These expressions are understood as maps acting on fields with their stated index symmetries; symmetrization in the contracted field slots is therefore implicit, and a representative differing by terms that vanish on symmetric fields defines the same map.

Appendix C: Tensor bases

Each kernel is expanded in the basis of all independent tensors with its index structure, built from η and p with the momentum dependence of Table II. The construction is the same for the four kernels. One enumerates the tensor monomials of the required degree in p , each a single product of η 's over the index slots (index positions of K) not carrying a momentum, times the appropriate powers of p , and symmetrizes each monomial under the index symmetries of the kernel. These symmetries, listed in Table II, generate a finite group $\mathbf{G}_K$ of permutations of the index slots of K . Applying every element of $\mathbf{G}_K$ to a monomial T produces a set of n distinct monomials $T_1, \dots, T_n$, the *orbit* of T ; the symmetrized monomial is the average over the orbit, $\mathcal{S}[T] = \frac{1}{n}\sum_{k=1}^n T_k$ (equivalently, the average of T over the

group). Two monomials give the same $\mathcal{S}[T]$ if and only if they lie in the same orbit, so the orbit averages form a basis of the invariant formal expressions, in one-to-one correspondence with the orbits, and any element of an orbit may serve as its representative. This is a candidate tensor basis; its independence as a set of tensors in a given dimension is checked below. The orbit sizes n add up to the number of monomials.

Tables V–VIII list, for each kernel, a representative monomial of every orbit and the number n of terms in the corresponding symmetrized sum. Indices are written in the order of the kernel’s slots; $p^2 \equiv p_\alpha p^\alpha$. For $K^{\phi\phi}$, for instance, element 4 of Table V is $\frac{1}{4}(p_\mu p_\rho \eta_{\nu\sigma} + p_\nu p_\rho \eta_{\mu\sigma} + p_\mu p_\sigma \eta_{\nu\rho} + p_\nu p_\sigma \eta_{\mu\rho})$.

Distinct orbits have disjoint monomial content, so the elements are linearly independent as formal expressions. They can nevertheless become dependent as tensors in low dimension, through identities that hold only for specific d . Independence for generic d is verified by computing the Gram matrix Γ_K of each basis, $(\Gamma_K)_{ij} = \mathcal{S}[T_i] \cdot \mathcal{S}[T_j]$ with all indices contracted, and checking that its determinant is not identically zero; the determinant is a polynomial in d whose zeros locate the dimensions where the basis degenerates. A nonzero determinant establishes linear independence in any signature. Since the entries of Γ_K depend on the signature only through d and p^2 , the dimension-dependent degeneracies can be identified at nonzero Euclidean momentum, where the contraction pairing is positive definite and a vanishing determinant does imply linear dependence; the resulting polynomial tensor relations hold in Lorentzian signature as well. For the bases of Tables V–VIII, up to positive numerical factors,

$$\det \Gamma_{\phi\phi} = (p^2)^8 (d-2)(d-1)^3(d+2)(d+4), \quad (\text{C1})$$

$$\det \Gamma_{\phi G} = (p^2)^6 (d-2)(d-1)^5(d+2)^5(d+4), \quad (\text{C2})$$

$$\det \Gamma_{GG} = d^5 (d-2)(d-1)^4(d+2)^4(d+4), \quad (\text{C3})$$

$$\det \Gamma_{\phi GG} = d^{16} (d-3)(d-2)^7(d-1)^{15}(d+1)^3(d+2)^{15}(d+4)^6(d+6). \quad (\text{C4})$$

All four are nonzero for $d \geq 4$ and $p^2 \neq 0$; the powers of p^2 reflect the momentum dependence of the first two bases and are unrelated to dimension-dependent relations. The zeros at $d = 1, 2$, and at $d = 3$ for $K^{\phi GG}$, are the low dimensions in which products of η ’s with many indices become linearly dependent (in $d = 3$, for instance, antisymmetrization over four indices vanishes).

Representative monomial n	
1 $p^2 \eta_{\mu\nu} \eta_{\rho\sigma}$	1
2 $p^2 \eta_{\mu\rho} \eta_{\nu\sigma}$	2
3 $p_\mu p_\nu \eta_{\rho\sigma}$	2
4 $p_\mu p_\rho \eta_{\nu\sigma}$	4

TABLE V. Basis of $K_{\mu\nu\rho\sigma}^{\phi\phi}(p)$: 4 elements, the orbits of the 9 monomials under the symmetry group of order 8.

Representative monomial n	
1 $p_\mu \eta_{\nu\lambda} \eta_{\rho\sigma}$	2
2 $p_\mu \eta_{\nu\rho} \eta_{\lambda\sigma}$	4
3 $p_\lambda \eta_{\mu\nu} \eta_{\rho\sigma}$	1
4 $p_\lambda \eta_{\mu\rho} \eta_{\nu\sigma}$	2
5 $p_\rho \eta_{\mu\nu} \eta_{\lambda\sigma}$	2
6 $p_\rho \eta_{\mu\lambda} \eta_{\nu\sigma}$	4

TABLE VI. Basis of $K_{\mu\nu\lambda\rho\sigma}^{\phi G}(p)$: 6 elements, the orbits of the 15 monomials under the symmetry group of order 4.

Appendix D: The six identities

In the notation of Section II, each identity of Table III states that a multilinear form in the fields and the gauge parameter vanishes identically. Writing P for the total momentum

Representative monomial n	
1 $\eta_{\lambda\mu}\eta_{\nu\rho}\eta_{\sigma\tau}$	4
2 $\eta_{\lambda\mu}\eta_{\nu\sigma}\eta_{\rho\tau}$	4
3 $\eta_{\lambda\rho}\eta_{\mu\nu}\eta_{\sigma\tau}$	1
4 $\eta_{\lambda\rho}\eta_{\mu\sigma}\eta_{\nu\tau}$	2
5 $\eta_{\lambda\sigma}\eta_{\mu\rho}\eta_{\nu\tau}$	4

TABLE VII. Basis of $K_{\lambda\mu\nu\rho\sigma\tau}^{GG}$: 5 elements, the orbits of the 15 monomials under the symmetry group of order 8.

Representative monomial n		Representative monomial n	
1 $\eta_{\mu\nu}\eta_{\lambda\alpha}\eta_{\beta\rho}\eta_{\sigma\tau}$	4	9 $\eta_{\mu\lambda}\eta_{\nu\rho}\eta_{\alpha\sigma}\eta_{\beta\tau}$	4
2 $\eta_{\mu\nu}\eta_{\lambda\alpha}\eta_{\beta\sigma}\eta_{\rho\tau}$	4	10 $\eta_{\mu\lambda}\eta_{\nu\sigma}\eta_{\alpha\beta}\eta_{\rho\tau}$	8
3 $\eta_{\mu\nu}\eta_{\lambda\rho}\eta_{\alpha\beta}\eta_{\sigma\tau}$	1	11 $\eta_{\mu\lambda}\eta_{\nu\sigma}\eta_{\alpha\rho}\eta_{\beta\tau}$	16
4 $\eta_{\mu\nu}\eta_{\lambda\rho}\eta_{\alpha\sigma}\eta_{\beta\tau}$	2	12 $\eta_{\mu\alpha}\eta_{\nu\beta}\eta_{\lambda\rho}\eta_{\sigma\tau}$	4
5 $\eta_{\mu\nu}\eta_{\lambda\sigma}\eta_{\alpha\rho}\eta_{\beta\tau}$	4	13 $\eta_{\mu\alpha}\eta_{\nu\beta}\eta_{\lambda\sigma}\eta_{\rho\tau}$	8
6 $\eta_{\mu\lambda}\eta_{\nu\alpha}\eta_{\beta\rho}\eta_{\sigma\tau}$	8	14 $\eta_{\mu\alpha}\eta_{\nu\sigma}\eta_{\lambda\beta}\eta_{\rho\tau}$	8
7 $\eta_{\mu\lambda}\eta_{\nu\alpha}\eta_{\beta\sigma}\eta_{\rho\tau}$	16	15 $\eta_{\mu\alpha}\eta_{\nu\sigma}\eta_{\lambda\rho}\eta_{\beta\tau}$	8
8 $\eta_{\mu\lambda}\eta_{\nu\rho}\eta_{\alpha\beta}\eta_{\sigma\tau}$	2	16 $\eta_{\mu\alpha}\eta_{\nu\sigma}\eta_{\lambda\tau}\eta_{\beta\rho}$	8

TABLE VIII. Basis of $K_{\mu\nu\lambda\alpha\beta\rho\sigma\tau}^{\phi GG}$: 16 elements, the orbits of the 105 monomials (one for each perfect matching of the eight index slots by four η 's) under the symmetry group of order 16.

($P = p_1 + p_2$ or $p_1 + p_2 + p_3$), the six forms are

$$\mathcal{W}^\phi : \quad 2K^{\phi\phi}(\phi_{-p}, D_{\phi^0}(p) \omega_p) + K^{\phi G}(\phi_{-p}, D_{G^0}(p) \omega_p) = 0, \quad (\text{D1})$$

$$\mathcal{W}^G : \quad K^{\phi G}(D_{\phi^0}(p) \omega_p, G_{-p}) + 2K^{GG}(G_{-p}, D_{G^0}(p) \omega_p) = 0, \quad (\text{D2})$$

$$\mathcal{W}^{31} : \quad K^{\phi\phi}(\phi_{-p_1}, D_{\phi^1}(p_1, P)(\phi_{-p_2}, \omega_P)) + (1 \leftrightarrow 2) = 0, \quad (\text{D3})$$

$$\begin{aligned} \mathcal{W}^{32} : \quad & 2K^{\phi GG}(\phi_{-p_2}, D_{G^0}(P) \omega_P, G_{-p_1}) + K^{\phi G}(D_{\phi^1}(p_1, P)(\phi_{-p_2}, \omega_P), G_{-p_1}) \\ & + K^{\phi G}(\phi_{-p_2}, D_{G^1}(p_2, P)(G_{-p_1}, \omega_P)) = 0, \end{aligned} \quad (\text{D4})$$

$$\begin{aligned} \mathcal{W}^{33} : \quad & K^{\phi GG}(D_{\phi^0}(P) \omega_P, G_{-p_1}, G_{-p_2}) + K^{GG}(G_{-p_1}, D_{G^1}(p_1, P)(G_{-p_2}, \omega_P)) + (1 \leftrightarrow 2) = 0, \\ & \quad \quad \quad (\text{D5}) \end{aligned}$$

$$\begin{aligned} \mathcal{W}^{34} : \quad & K^{\phi GG}(\phi_{-p_3}, G_{-p_1}, D_{G^1}(p_1 + p_3, P)(G_{-p_2}, \omega_P)) + (1 \leftrightarrow 2) \\ & + K^{\phi GG}(D_{\phi^1}(p_1 + p_2, P)(\phi_{-p_3}, \omega_P), G_{-p_1}, G_{-p_2}) = 0, \end{aligned} \quad (\text{D6})$$

where $(1 \leftrightarrow 2)$ adds the immediately preceding term with the two field arguments of the same type exchanged, that is, with $p_1 \leftrightarrow p_2$ and, in the component form below, with their index groups exchanged as well; in $\mathcal{W}^{33}$ it applies only to the K^{GG} term. The factors of two count the two equal contributions of a kernel that is quadratic in the field being varied. Each term is one kernel of Eq. (3.1) with one argument replaced by the appropriate piece of the transformation (2.5), the momenta being fixed by momentum conservation. Since the fields are arbitrary within their symmetry classes, the coefficient tensor of $\phi_{-p_1} \cdots \omega_P$ in each form, projected onto the index symmetries of the field slots, vanishes. With all indices

displayed, and in terms of the tensors of Appendix B, these coefficient tensors are

$$\mathcal{W}_{\rho\sigma\alpha}^\phi(p) \equiv 2D_{\phi^0}{}^{\mu\nu}{}_\alpha(p)K_{\mu\nu\rho\sigma}^{\phi\phi}(p) + D_{G^0}{}^\lambda{}_{\mu\nu\alpha}(p)K_{\rho\sigma\lambda}^{\phi G}{}^{\mu\nu}(p) = 0, \quad (\text{D7})$$

$$\mathcal{W}_{\rho\sigma\tau\alpha}^G(p) \equiv D_{\phi^0}{}^{\mu\nu}{}_\alpha(p)K_{\mu\nu\rho\sigma\tau}^{\phi G}(-p) + 2D_{G^0}{}^\lambda{}_{\mu\nu\alpha}(p)K_\lambda^{GG\mu\nu}{}_{\rho\sigma\tau} = 0, \quad (\text{D8})$$

$$\begin{aligned} \mathcal{W}_{\mu_1\nu_1\mu_2\nu_2\rho}^{31}(p_1, p_2) &\equiv K_{\mu_1\nu_1\alpha\beta}^{\phi\phi}(p_1)D_{\phi^1}{}^{\alpha\beta}{}_{\mu_2\nu_2\rho}(p_1, p_1 + p_2) \\ &\quad + K_{\mu_2\nu_2\alpha\beta}^{\phi\phi}(p_2)D_{\phi^1}{}^{\alpha\beta}{}_{\mu_1\nu_1\rho}(p_2, p_1 + p_2) = 0, \end{aligned} \quad (\text{D9})$$

$$\begin{aligned} \mathcal{W}_{\lambda_1\mu_1\nu_1\mu_2\nu_2\rho}^{32}(p_1, p_2) &\equiv 2K_{\mu_2\nu_2\lambda\mu\nu\lambda_1\mu_1\nu_1}^{\phi GG}D_{G^0}{}^{\lambda\mu\nu}{}_\rho(p_1 + p_2) \\ &\quad + K_{\mu\nu\lambda_1\mu_1\nu_1}^{\phi G}(-p_1)D_{\phi^1}{}^{\mu\nu}{}_{\mu_2\nu_2\rho}(p_1, p_1 + p_2) \\ &\quad + K_{\mu_2\nu_2\lambda}^{\phi G}{}^{\mu\nu}(p_2)D_{G^1}{}^\lambda{}_{\mu\nu\lambda_1\mu_1\nu_1\rho}(p_2, p_1 + p_2) = 0, \end{aligned} \quad (\text{D10})$$

$$\begin{aligned} \mathcal{W}_{\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2\rho}^{33}(p_1, p_2) &\equiv K_{\mu\nu\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2}^{\phi GG}D_{\phi^0}{}^{\mu\nu}{}_\rho(p_1 + p_2) \\ &\quad + K_{\lambda\mu\nu\lambda_1\mu_1\nu_1}^{GG}D_{G^1}{}^{\lambda\mu\nu}{}_{\lambda_2\mu_2\nu_2\rho}(p_1, p_1 + p_2) \\ &\quad + K_{\lambda\mu\nu\lambda_2\mu_2\nu_2}^{GG}D_{G^1}{}^{\lambda\mu\nu}{}_{\lambda_1\mu_1\nu_1\rho}(p_2, p_1 + p_2) = 0, \end{aligned} \quad (\text{D11})$$

$$\begin{aligned} \mathcal{W}_{\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2\mu_3\nu_3\rho}^{34}(p_1, p_2, p_3) &\equiv \\ &\quad K_{\mu_3\nu_3\lambda\mu\nu\lambda_1\mu_1\nu_1}^{\phi GG}D_{G^1}{}^{\lambda\mu\nu}{}_{\lambda_2\mu_2\nu_2\rho}(p_1 + p_3, P) \\ &\quad + K_{\mu_3\nu_3\lambda\mu\nu\lambda_2\mu_2\nu_2}^{\phi GG}D_{G^1}{}^{\lambda\mu\nu}{}_{\lambda_1\mu_1\nu_1\rho}(p_2 + p_3, P) \\ &\quad + K_{\mu\nu\lambda_1\mu_1\nu_1\lambda_2\mu_2\nu_2}^{\phi GG}D_{\phi^1}{}^{\mu\nu}{}_{\mu_3\nu_3\rho}(p_1 + p_2, P) = 0, \end{aligned} \quad (\text{D12})$$

The identities above retain their generic momentum dependence. For the projection bases of the cubic identities the independent momenta are identified, $p_1 = p_2 = p$, and also $p_3 = p$ in $\mathcal{W}^{34}$, as explained in Section IV. The projection bases of Table III are generated by the procedure of Appendix C, from the monomials with the index structure and degree in p of each identity, averaged over its symmetry group: the symmetries of the kernel slots involved and, for $\mathcal{W}^{31}$, $\mathcal{W}^{33}$ and $\mathcal{W}^{34}$, the exchange of the two identical fields. Table IX lists the three smallest ones as an illustration; the remaining dimensions, 23, 20 and 95, make an explicit listing unhelpful, and those bases are generated by the code of Appendix F. Their Gram determinants are, up to numerical factors, $(p^2)^9(d-1)^2$, $(p^2)^{12}(d-2)(d-1)^5(d+2)(d+4)$ and $(p^2)^{21}(d-2)^2(d-1)^6(d+1)(d+4)(d+6)$ for the three bases of Table IX, and for $p^2 \neq 0$ those of the bases of $\mathcal{W}^{32}$ and $\mathcal{W}^{33}$ vanish, among positive integer dimensions, only at $d = 1, 2, 3$; all five are therefore nondegenerate for $d \geq 4$ and $p^2 \neq 0$. The basis of $\mathcal{W}^{34}$ becomes degenerate at $d = 4$. This does not obstruct the determination, since no inversion of that Gram matrix is needed: the other identities fix the kernels, $\mathcal{W}^{34}$ is then found to be

satisfied identically by substitution, and its vanishing also follows from the identification of the solution with the invariant Goldberg–Palatini action.

Representative monomial n		Representative monomial n	
$\mathcal{W}_{\rho\sigma\alpha}^{\phi}(p)$		$\mathcal{W}_{\mu_1\nu_1\mu_2\nu_2\rho}^{31}(p, p)$	
1	$p_{\rho}p_{\sigma}p_{\alpha}$	1	$p_{\mu_1}p_{\nu_1}p_{\rho}\eta_{\mu_2\nu_2}$
2	$p^2 p_{\alpha}\eta_{\rho\sigma}$	2	$p_{\mu_1}p_{\nu_1}p_{\mu_2}\eta_{\nu_2\rho}$
3	$p^2 p_{\rho}\eta_{\sigma\alpha}$	3	$p_{\mu_1}p_{\mu_2}p_{\rho}\eta_{\nu_1\nu_2}$
$\mathcal{W}_{\rho\sigma\tau\alpha}^G(p)$		4	$p^2 p_{\rho}\eta_{\mu_1\nu_1}\eta_{\mu_2\nu_2}$
1	$p_{\rho}p_{\alpha}\eta_{\sigma\tau}$	5	$p^2 p_{\rho}\eta_{\mu_1\mu_2}\eta_{\nu_1\nu_2}$
2	$p_{\sigma}p_{\tau}\eta_{\rho\alpha}$	6	$p^2 p_{\mu_1}\eta_{\nu_1\rho}\eta_{\mu_2\nu_2}$
3	$p_{\rho}p_{\sigma}\eta_{\tau\alpha}$	7	$p^2 p_{\mu_1}\eta_{\nu_1\mu_2}\eta_{\nu_2\rho}$
4	$p_{\sigma}p_{\alpha}\eta_{\rho\tau}$		
5	$p^2 \eta_{\rho\alpha}\eta_{\sigma\tau}$		
6	$p^2 \eta_{\rho\sigma}\eta_{\tau\alpha}$		

TABLE IX. The three smallest projection bases, in the notation of Appendix C: 3, 6 and 7 elements, the orbits of 4, 9 and 25 monomials under symmetry groups of order 2, 2 and 8. For $\mathcal{W}^{31}$ the two momenta are identified, $p_1 = p_2 = p$, and the symmetry group includes the exchange $(\mu_1\nu_1) \leftrightarrow (\mu_2\nu_2)$.

Appendix E: Explicit kernels

Up to the overall constant, the solution of Table IV is, in terms of the basis elements $\mathcal{S}[T]$ of Appendix C (the average of the monomial T over its orbit), and with all indices of

the kernels lowered, to be raised with $\eta^{\mu\nu}$ as required by the contractions with the fields,

$$K_{\mu\nu\rho\sigma}^{\phi\phi}(p) = 0, \quad (\text{E1})$$

$$K_{\mu\nu\lambda\rho\sigma}^{\phi G}(p) = -i \mathcal{S}[p_\lambda \eta_{\mu\rho} \eta_{\nu\sigma}] = -\frac{i}{2} p_\lambda (\eta_{\mu\rho} \eta_{\nu\sigma} + \eta_{\mu\sigma} \eta_{\nu\rho}), \quad (\text{E2})$$

$$\begin{aligned} K_{\lambda\mu\nu\rho\sigma\tau}^{GG} &= -\mathcal{S}[\eta_{\lambda\sigma} \eta_{\mu\rho} \eta_{\nu\tau}] + \frac{1}{d-1} \mathcal{S}[\eta_{\lambda\mu} \eta_{\nu\sigma} \eta_{\rho\tau}] \\ &= \frac{1}{4} \left[-\eta_{\lambda(\sigma} \eta_{\tau)(\mu} \eta_{\nu)\rho} + \frac{1}{d-1} \eta_{\lambda(\mu} \eta_{\nu)(\sigma} \eta_{\tau)\rho} \right], \end{aligned} \quad (\text{E3})$$

$$K_{\mu\nu\lambda\alpha\beta\rho\sigma\tau}^{\phi GG} = -\mathcal{S}[\eta_{\mu\alpha} \eta_{\nu\sigma} \eta_{\lambda\tau} \eta_{\beta\rho}] + \frac{1}{d-1} \mathcal{S}[\eta_{\mu\alpha} \eta_{\nu\sigma} \eta_{\lambda\beta} \eta_{\rho\tau}], \quad (\text{E4})$$

that is, element 4 of Table VI, elements 5 and 2 of Table VII, and elements 16 and 14 of Table VIII; the latter two basis elements contain eight terms each. In the GG and ϕGG kernels the first structure contracts the connection index of one G with a lower index of the other, and the second is the product of the two traces $g_\mu \equiv G^\sigma_{\mu\sigma}$. Contracted with the fields, in the notation of Section II, these give

$$K^{\phi G}(\phi_{-p}, G_p) = -ip_\lambda \tilde{\phi}^{\mu\nu}(-p) \tilde{G}^\lambda_{\mu\nu}(p), \quad (\text{E5})$$

$$K^{GG}(G_{-p}, G_p) = \eta^{\mu\nu} \left(-\tilde{G}^\lambda_{\mu\sigma}(-p) \tilde{G}^\sigma_{\nu\lambda}(p) + \frac{1}{d-1} \tilde{G}^\sigma_{\mu\sigma}(-p) \tilde{G}^\rho_{\nu\rho}(p) \right), \quad (\text{E6})$$

$$K^{\phi GG}(\phi_{p+q}, G_{-p}, G_{-q}) = \tilde{\phi}^{\mu\nu}(p+q) \left(-\tilde{G}^\lambda_{\mu\sigma}(-p) \tilde{G}^\sigma_{\nu\lambda}(-q) + \frac{1}{d-1} \tilde{G}^\sigma_{\mu\sigma}(-p) \tilde{G}^\rho_{\nu\rho}(-q) \right), \quad (\text{E7})$$

whose inverse Fourier transform yields Eq. (5.1), after an integration by parts in the ϕG term.

Appendix F: Symbolic implementation and reproducibility

In the first-order bootstrap implementation the coupling is set to $\kappa = 1$. Since κ enters the construction only as the overall factor of S_3 in Eq. (3.1) and of $\delta^{(1)}$ in Eq. (2.5), this affects neither the kernels of Appendix E nor the identities of Appendix D; it is restored in the results by dimensional analysis, as in Section III. The second-order comparison keeps κ symbolic throughout.

a. Files. The calculations are organized in the following Mathematica [15] notebooks and data files, kept together in one directory.

1. `Goldberg_Palatini_Bootstrap_en.nb` (FeynCalc [17]): the first-order determination. It generates the tensor bases of the four kernels and of the six identities (Appen-

dices [C](#) and [D](#)) and the Gram determinants of the kernel bases and of the five smallest projection bases (the rank test of the 95-element basis of $\mathcal{W}^{34}$ at $d = 4$ is an optional diagnostic, not part of the main flow), builds the nonlinear identities with the field momenta identified, solves the linear system in the sequence of Table [IV](#), checks the resulting kernels against Eqs. (3.5)–(3.6) of Ref. [\[10\]](#), verifies the complete cubic kernel against $\frac{1}{2}M(\phi)$ (cell tagged `FullCubicKernelCheck`), and exports the normalized kernels.

2. `Goldberg_Palatini_Bootstrap_kernels.wl`: the exported kernels K^{GG} and $K^{\phi GG}$ in a package-independent slot representation, with the conventions (dimension symbol, symmetries, normalization $c_{GG} = 1/(d-1)$, Fourier sign); it is data, not an executable script.
3. `Goldberg_Palatini_Cubic_CLM.nb` (`xAct/xTensor`, `xTras` and `xPerm` [\[16\]](#)): the second-order comparison of Section [VI](#). It imports the data file without `FeynCalc`, converts it to `xAct` tensors, derives M_0^{-1} by solving its defining identity in a five-element basis of metric structures and checks both inverse relations (`inverseChecks`), substitutes G_0 back into $M_0 G_0 = \partial\phi$ as a test of index placement and normalization (`g0Check`), and evaluates the residuals of Eqs. (6.4)–(6.5) (`clmQuadraticCheck`, `quadraticBoundaryCheck`, `clmCubicResidual`); `allChecksPassed` only aggregates these residuals. The notebook uses the metric-compatible derivative of the flat background, `CD`, which in Cartesian coordinates is the partial derivative of the text; the dimension is kept symbolic, and the pole of M_0^{-1} at $d = 2$ is separate from the domain $d \geq 4$ of the bases.

A `README` gives the evaluation order (the first notebook and its export cell first; the comparison notebook in a fresh kernel), the required packages and versions, the expected result of every check, the conventions of the comparison, and the correspondence between the notebooks and the equations and tables of this paper.

[Procedure to make the code executable by anyone, to be completed before submission (this note is to be removed from the final version, leaving only the files, requirements and instructions actually available): (i) assemble the public bundle with the files above, the `README` (already written) and the drivers of item (ii), without backup copies or private paths (all input/output relative to the notebook or script directory; the notebooks may keep

their formatting and front-end output); (ii) write one plain-text driver per notebook (`.wls`) that runs under `wolframscript` without a front end, prints each check and its residual, and exits with an error if any residual is nonzero; (iii) install the Wolfram Engine on a clean machine under its free license, which requires an account, activation and acceptance of its terms, together with `FeynCalc`, `xAct`, `xTras` and `xPerm`, and run the drivers there; record the exact versions and the outputs in the README; (iv) publish the bundle in a public repository with a tagged release and a Zenodo DOI, and cite it here and in a data-availability statement, not in the abstract; (v) only then state that the calculations can be reproduced without a paid Mathematica license, for the uses permitted by the terms of the free Wolfram Engine license. The Wolfram Engine is free but not license-free, and notebooks need either the Mathematica front end or the drivers of item (ii). A route with no Wolfram license at all would require re-implementing the checks in free software (the bases and Gram determinants already exist as Python/SymPy scripts); whether to include such a re-implementation is left open.]

The notebooks and data files will be made available at [repository URL and Zenodo DOI].

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- [1] R. P. Feynman, F. B. Morinigo, and W. G. Wagner, *Feynman Lectures on Gravitation*, edited by B. Hatfield (Addison-Wesley, Reading, MA, 1995).
 - [2] S. Deser, *Gen. Rel. Grav.* **1**, 9 (1970), [arXiv:gr-qc/0411023](#).
 - [3] T. Padmanabhan, *Int. J. Mod. Phys. D* **17**, 367 (2008), [arXiv:gr-qc/0409089](#).
 - [4] S. Deser, *Gen. Rel. Grav.* **42**, 641 (2010), [arXiv:0910.2975](#).
 - [5] F. T. Brandt and M. R. Elias-Filho, *Phys. Rev. D* **74**, 067704 (2006), [arXiv:hep-th/0609106](#).
 - [6] F. T. Brandt and J. Frenkel, *Phys. Rev. D* **53**, 911 (1996), [arXiv:hep-th/9505114](#).
 - [7] J. N. Goldberg, *Phys. Rev.* **111**, 315 (1958).
 - [8] A. Palatini, *Rend. Circ. Mat. Palermo* **43**, 203 (1919).
 - [9] M. Ferraris, M. Francaviglia, and C. Reina, *Gen. Rel. Grav.* **14**, 243 (1982).
 - [10] F. T. Brandt and D. G. C. McKeon, *Phys. Rev. D* **93**, 105037 (2016), [arXiv:1601.04944](#).
 - [11] A. Delhom, G. García-Moreno, M. Hohmann, A. Jiménez-Cano, and T. S. Koivisto, *JCAP* **12** (12), 006, [arXiv:2211.13056](#).
 - [12] R. M. Wald, *Phys. Rev. D* **33**, 3613 (1986).

- [13] F. T. Brandt and D. G. C. McKeon, [Phys. Rev. D **91**, 105006 \(2015\)](#), [arXiv:1503.02598](#).
- [14] D. M. Capper, G. Leibbrandt, and M. Ramón Medrano, [Phys. Rev. D **8**, 4320 \(1973\)](#).
- [15] Wolfram Research, Inc., Mathematica, version 15.0, Champaign, IL (2026).
- [16] J. M. Martín-García, [Comput. Phys. Commun. **179**, 597 \(2008\)](#).
- [17] V. Shtabovenko, R. Mertig, and F. Orellana, [Comput. Phys. Commun. **256**, 107478 \(2020\)](#).